XTREMAX
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Hi !

Thank you for your interest in joining Xtremax as :

Cloud Engineer

Cloud Engineer Test

Below is a cloud engineer study case to peel off your insight about Cloud Engineering.

General Info

In this test you are **required to submit the report within 4 days after receiving the test case**. Please follow the report format to answer this test in the last section of this document. If you have any questions don't hesitate to contact Xtremax Recruiter.

Instruction

Use Case:

PT Yakin Bisa Lolos is planning to deploy a two-tier web application for school websites on the AWS cloud infrastructure. The application serves a WordPress website. The architecture includes an Amazon CloudFront distribution, an Application Load Balancer (ALB), Amazon EC2 instances, and an RDS MySQL database. As a Cloud Engineer, your task is to utilize Terraform to provision all the required resources and configurations securely.

A. Requirements:

- Use Terraform for Provisioning:
  Utilize Terraform to automate the provisioning of all resources, ensuring a repeatable and version-controlled deployment of the infrastructure.
- Provision and Configure Resources:
  i. Set up a three-tier web application architecture, including:
    1. Amazon CloudFront:
       Configure CloudFront as a Content Delivery Network (CDN) to optimize the delivery of public connections
    2. Application Load Balancer (ALB):
       Provision an ALB to distribute incoming traffic across multiple EC2 instances for high availability.
    3. Amazon EC2 Instances:
       a. Deploy EC2 instances running the WordPress image on Autoscaling Group
       b. Set up autoscaling with a minimum of one server and a maximum of 4 servers.
       c. Configure upscaling when CPU utilization exceeds 80% for 30 minutes and downscaling when CPU utilization drops below 40% for 30 minutes.
    4. Amazon RDS MySQL Database:
       Set up an RDS MySQL database to store the website data securely.
- Security Considerations:
  i. Implement necessary security measures for all resources:
    1. Enable HTTPS for the CloudFront distribution and ALB to encrypt data in transit.
    2. Configure security groups and network ACLs to restrict incoming and outgoing traffic to the EC2 instances and RDS database.
    3. Use IAM roles and policies to grant the minimum required permissions to resources.

Deliverables

For this test, you need to submit these deliverables:

- Terraform scripts for provisioning the entire infrastructure.
- Documentation describing the architecture, configurations, and security measures implemented.
- Test plan and results to ensure the correct functioning and security of the deployed resources.

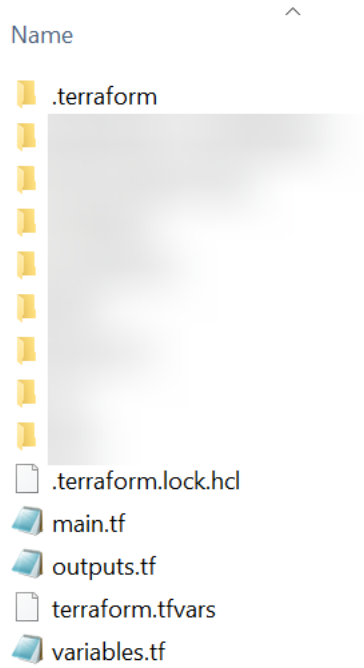
Grading

Candidates will be evaluated based on the following:

- Completeness and correctness of the Terraform scripts.
- Adherence to best practices in infrastructure provisioning.
- Effectiveness of security measures implemented.
- Clarity and completeness of documentation.

Document Format

❖ Please create the terraform files and put inside zip:



❖ Please give example of system diagram that build this web school project:

Example:
<Insert Your Diagram Here>

❖ Please give example of module structure diagram that build this web school project:

Example:

```
<root>
<module 1>
  <sub module>
<module 2>
. . .
```

❖ Please describe each module alongside its output and input:

Example:

```
#main.tf at the root
Combining all module's folder that build a Web School Project
#terraform.tfvars
Consists of Input required for all modules
#outputs.tf at the root
Displaying final output from a web school project is deployed
#variables.tf at the root
Declaration of variable that is used on main.tf at the root
#Module 1
. . .
Importing module 2 . . .
Getting output from module 2 . . .
#Module 2
. . .
```

FAQ

1. Do I need to provision real resources in AWS?

   Not necessarily needed, you can provide the terraform plan result as proof.

2. Can I submit the documentation separately?

   No, please send all deliverables in 1 email.

3. How long does it take to review my submission?

   It takes around 1-2 weeks to review, and the recruiter will inform you regarding the result.

GOOD LUCK!